

Experiment 9: Comparison of Linear and Polynomial Regression

Aim

To implement and compare Linear Regression and Polynomial Regression using Python.

Algorithm

Step 1: Import the required libraries.

Step 2: Create a sample dataset.

Step 3: Train the Linear Regression model using the dataset.

Step 4: Transform the input features into polynomial features.

Step 5: Train the Polynomial Regression model.

Step 6: Predict the output using both models.

Step 7: Calculate and compare the R^2 scores of both models.

Step 8: Display the results.

Program

```
#Experiment 9: Comparison of Linear and Polynomial Regression
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score

X = np.array([[1],[2],[3],[4],[5],[6]])
y = np.array([1,4,9,16,25,36])

linear = LinearRegression()
linear.fit(X, y)

poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = PolynomialRegression()
poly_model.fit(X_poly, y)

linear_pred = linear.predict(X)
poly_pred = poly_model.predict(X_poly)

print("Linear Regression R2 Score:", r2_score(y, linear_pred))
print("Polynomial Regression R2 Score:", r2_score(y, poly_pred))
```

Output

```
*** Linear Regression R2 Score: 0.9582790091264668  
Polynomial Regression R2 Score: 1.0
```

Result

Thus, Linear Regression and Polynomial Regression were successfully implemented and compared. Polynomial Regression provided better accuracy than Linear Regression for the given nonlinear dataset.